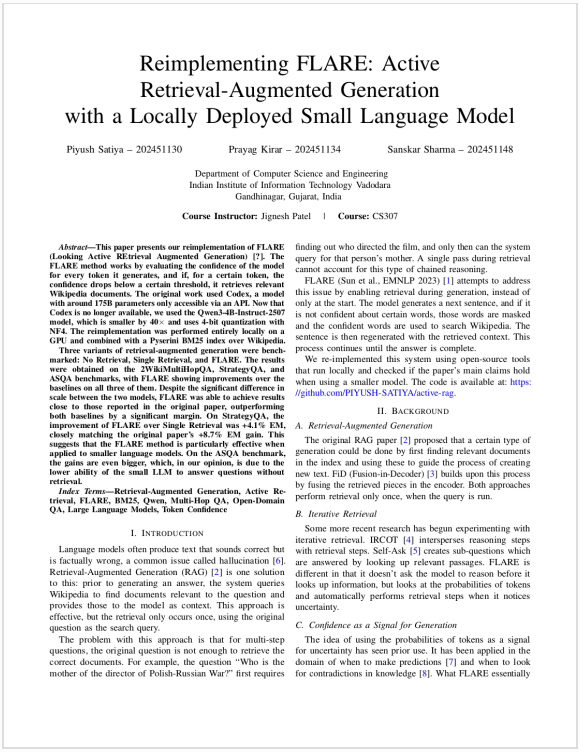
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Reimplementing FLARE: Active Retrieval-Augmented Generation with a Locally Deployed Small Language Model

Piyush Satiya – 202451130 Prayag Kirar – 202451134 Sanskar Sharma – 202451148

Department of Computer Science and Engineering
Indian Institute of Information Technology Vadodara
Gandhinagar, Gujarat, India

Course Instructor: Jignesh Patel | **Course:** CS307

***Abstract*—This paper presents our reimplementation of FLARE (Looking Active REtrieval Augmented Generation) [?]. The FLARE method works by evaluating the confidence of the model for every token it generates, and if, for a certain token, the confidence drops below a certain threshold, it retrieves relevant Wikipedia documents. The original work used Codex, a model with around 175B parameters only accessible via an API. Now that Codex is no longer available, we used the Qwen3-4B-Instruct-2507 model, which is smaller by 40× and uses 4-bit quantization with NF4. The reimplementation was performed entirely locally on a GPU and combined with a Pyserini BM25 index over Wikipedia.**

Three variants of retrieval-augmented generation were benchmarked: No Retrieval, Single Retrieval, and FLARE. The results were obtained on the 2WikiMultiHopQA, StrategyQA, and ASQA benchmarks, with FLARE showing improvements over the baselines on all three of them. Despite the significant difference in scale between the two models, FLARE was able to achieve results close to those reported in the original paper, outperforming both baselines by a significant margin. On StrategyQA, the improvement of FLARE over Single Retrieval was +4.1% EM, closely matching the original paper's +8.7% EM gain. This suggests that the FLARE method is particularly effective when applied to smaller language models. On the ASQA benchmark, the gains are even bigger, which, in our opinion, is due to the lower ability of the small LLM to answer questions without retrieval.

***Index Terms*—Retrieval-Augmented Generation, Active Retrieval, FLARE, BM25, Qwen, Multi-Hop QA, Open-Domain QA, Large Language Models, Token Confidence**

I. Introduction

Language models often produce text that sounds correct but is factually wrong, a common issue called hallucination [6]. Retrieval-Augmented Generation (RAG) [2] is one solution to this: prior to generating an answer, the system queries Wikipedia to find documents relevant to the question and provides those to the model as context. This approach is effective, but the retrieval only occurs once, using the original question as the search query.

The problem with this approach is that for multi-step questions, the original question is not enough to retrieve the correct documents. For example, the question "Who is the mother of the director of Polish-Russian War?" first requires finding out who directed the film, and only then can the system query for that person's mother. A single pass during retrieval cannot account for this type of chained reasoning.

FLARE (Sun et al., EMNLP 2023) [1] attempts to address this issue by enabling retrieval during generation, instead of only at the start. The model generates a next sentence, and if it is not confident about certain words, those words are masked and the confident words are used to search Wikipedia. The sentence is then regenerated with the retrieved context. This process continues until the answer is complete.

We re-implemented this system using open-source tools that run locally and checked if the paper's main claims hold when using a smaller model. The code is available at: https://github.com/PIYUSH-SATIYA/active-rag.

II. Background

A. Retrieval-Augmented Generation

The original RAG paper [2] proposed that a certain type of generation could be done by first finding relevant documents in the index and using these to guide the process of creating new text. FiD (Fusion-in-Decoder) [3] builds upon this process by fusing the retrieved pieces in the encoder. Both approaches perform retrieval only once, when the query is run.

B. Iterative Retrieval

Some more recent research has begun experimenting with iterative retrieval. IRCOT [4] intersperses reasoning steps with retrieval steps. Self-Ask [5] creates sub-questions which are answered by looking up relevant passages. FLARE is different in that it doesn't ask the model to reason before it looks up information, but looks at the probabilities of tokens and automatically performs retrieval steps when it notices uncertainty.

C. Confidence as a Signal for Generation

The idea of using the probabilities of tokens as a signal for uncertainty has seen prior use. It has been applied in the domain of when to make predictions [7] and when to look for contradictions in knowledge [8]. What FLARE essentially

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